



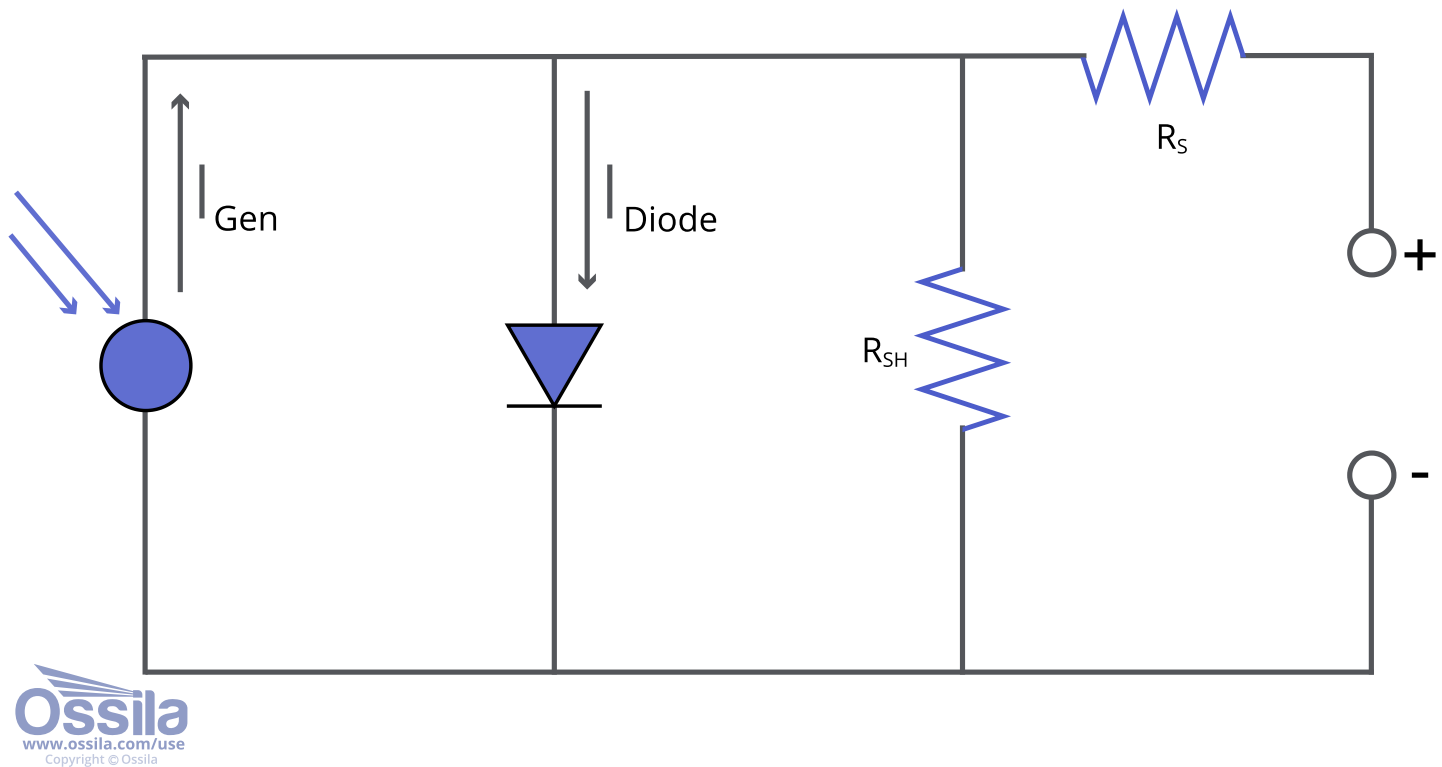
Equivalent Circuit of Solar Cell



An equivalent circuit model presents a theoretical circuit diagram, which captures the electrical characteristics of a device. It is important to note the components illustrated in the model are not physically present in the devices themselves. Instead, these models serve to help us visualize and simplify calculations related to the cell's behavior. These models are invaluable for understanding fundamental device physics, explaining specific phenomena, and aiding in the design of more efficient devices.

Solar Cell Equivalent Circuit

The equivalent circuit of a solar cell consists of an ideal current generator connected in parallel with a diode, both of which are connected to a load. The generated current is directly proportional to light intensity. This highlights how important it is to accurately replicate the solar spectrum when testing solar cells, and why a **solar simulator** is an indispensable piece of equipment in this context. Although the amount of current produced varies with light intensity, there are other limitations in solar cells which cap their efficiency. These limitations are represented by the other components in the circuit.



Equivalent circuit diagram of a solar cell

Parallel to this ideal current generator is a diode. The power that can be extracted from a device (P) is equal to current (I) times by voltage (V):

$$P = V \times I$$

If the resistance across the load surpasses that of the diode, the diode will draw current, increasing the potential difference between the terminals, but diminishing the current directed through the load. Alternatively, if the diode's resistance is greater than the load's, electrons easily flow through the load, leading to a higher current. However, the potential difference between the terminals will be relatively low. This illustrates a core constraint with solar cells: optimizing current often means compromising on voltage. There is a sweet spot, the maximum power point, where both voltage and current are optimized, maximizing power output.

Additionally, you can represent device losses using equivalent circuit diagrams. In the above ideal circuit diagram of a solar cell, there are components which represent series resistance and shunt resistance. Shunt resistance accounts for all losses that result in electrons traveling straight between the terminals, such as shorts in the device. It is therefore represented by a resistor running parallel to the ideal current generator and you should aim to increase shunt resistance as much as possible. This means you should do everything you can to ensure your terminals remain separated i.e., no pinholes or defects.

The other component in the diagram represents series resistance, which accounts for all current losses due to poor charge transfer between or within layers of your device. In the equivalent circuit diagram, this is depicted as a resistor in series with the ideal current generator. You should do everything you can to lower series resistance in order to allow seamless electron movement through the device.

Defining Equations and Metrics

From this ideal circuit diagram, we can extract equations to describe and model solar cells. This also helps us define some of the most important metrics we use to describe solar cells. In its simplest form, we can describe

current through the load as the amount of current generated minus the current that flows through the diodes and the current lost to shunt resistance.

$$I = I_{Gen} - I_{Diode} - I_{Sh}$$

where I is current extracted, I_{Gen} is the generated current, I_{Diode} is diode current, and I_{SH} is current lost to shunt resistance.

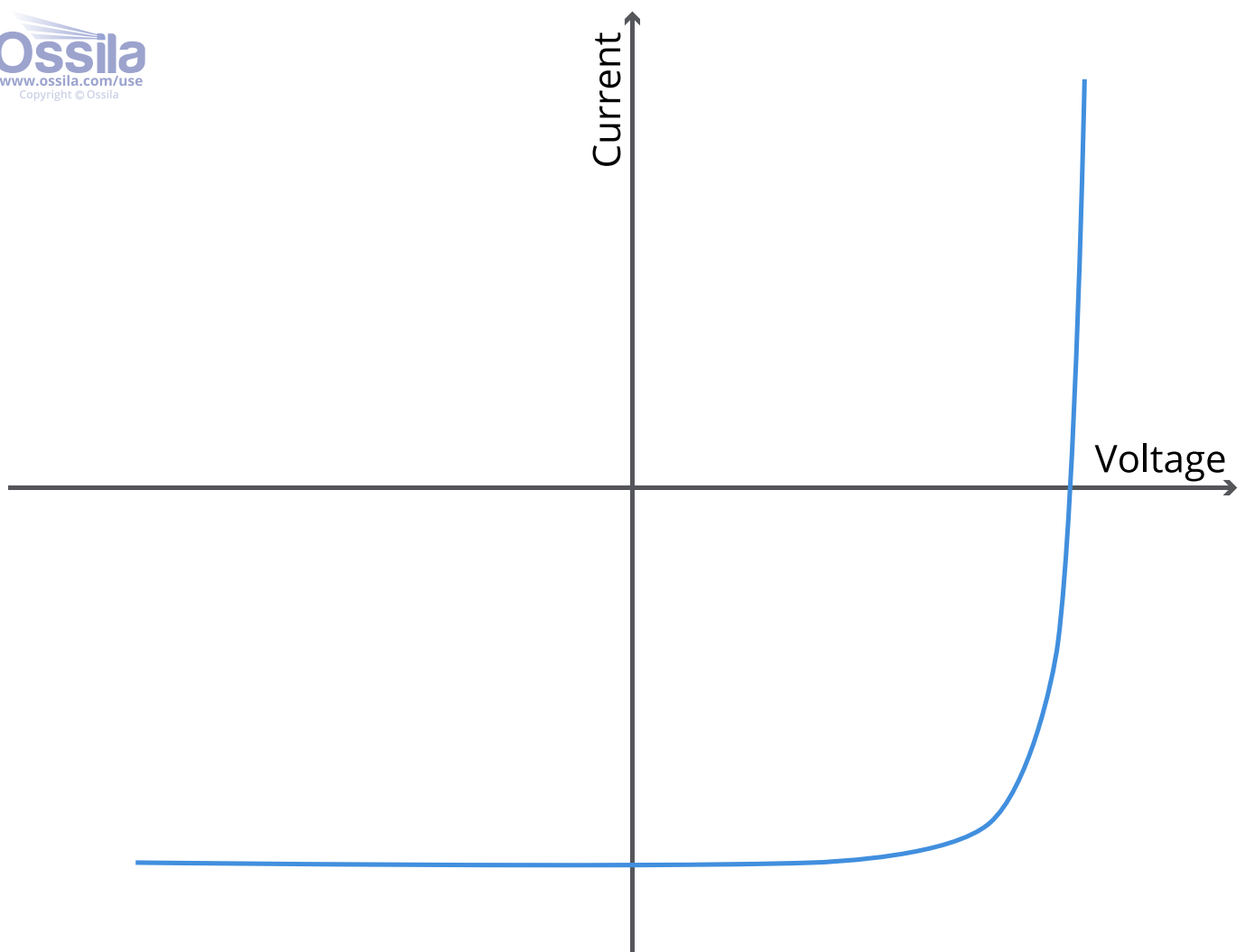
The ideal diode equation I_D is:

$$I_D = I_0 \left\{ \exp \left[\frac{qV}{nkT} \right] - 1 \right\}$$

Where I_0 is the reverse saturation current, n is the diode ideality factor, q is the charge constant, k is the Boltzmann constant, and T is absolute temperature. Therefore, the overall current equation can be written as:

$$I = I_{Gen} - I_0 \left\{ \exp \left[\frac{qV}{nkT} \right] - 1 \right\} - I_{Sh}$$

This equation gives us the characteristic current-voltage graph shape we see for solar cells.



I-V curve of a solar cell

We can also express this equation in terms of current density, J , where:

$$J = \frac{I}{Area}$$

Here, area refers to device area and I is the measured current. This allows us to define a current equation in terms of J.

$$J = J_{Gen} - J_0 \left\{ \exp \left[\frac{qV}{nkT} \right] - 1 \right\} - J_{Sh}$$

We can use this equation to express important device metrics. For the following equations, we will assume that shunt resistance is sufficiently high so that we can ignore the J_{SH} term. We can define $V=0$ in the above equation to find an expression for the short-circuit current J_{SC} .

$$J_{SC} \approx J_{Gen}$$

By doing this, we find that the current generated is approximately equal to J_{SC} . If this is true, we can re-express the above equation in terms of J_{SC} , and J_{Dark} , i.e., the current that will run through the solar cell in the absence of illumination.

$$J = J_{SC} - J_{Dark} = J_{SC} - J_0 \left\{ \exp \left[\frac{qV}{nkT} \right] - 1 \right\}$$

Alternatively, if we define $J=0$, we can find an expression for V_{OC} .

$$V_{OC} = \frac{nkT}{q} \ln \left(\frac{J_{Gen}}{J_0} + 1 \right)$$

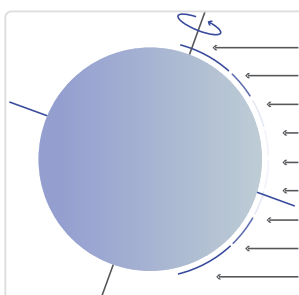
Understanding the equivalent circuit of a solar cell is more than just a theoretical exercise; it is a bridge between conceptual understanding and practical application. The equivalent circuit of a solar cell can help us visualize and explain the behavior of solar cells, providing vital information to help us make them more efficient.

Solar Simulator

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Solar Cell Testing & Characterization



One main application of solar simulators is to test solar cell devices and modules. To characterize how solar cells will perform in the real world, it is vital that you use a solar source that mimics the sun's spectrum well. You could of course use actual sunlight, b...

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